

STANES SCHOOL

1044, Avinashi Road, Coimbatore – 18.

UNIT TEST – 1 2026-2027

Class : XII
Subject : Computer Science

Duration : 2 Hours
Max. Marks : 50

SECTION A (17 × 1 = 17 Marks)

1. State whether the following statement is **True** or **False**:

The output of the following code will be 5.0

```
import math
print(math.sqrt(25))
```

2. Which of the following is a mutable data type in Python?

- a) Tuple b) String c) List d) None

3. What will be the output of the following code?

```
print(15 > 10 and 20 < 15 or not 5 > 2)
```

- a) True b) False c) Error d) No output

4. Which of the following is not a Python keyword?

- a) def b) elif c) func d) while

5. What will be the output of the following Python code?

```
str = "Programming"
print(str[-4:-1])
```

- a) min b) ing c) ming d) ami

6. Write the output of the following Python code:

```
for k in range(2, 15, 3):
    print(k, end=" ")
```

7. What will be the output of the following Python statement?

```
print(12 + 3 ** 2 - 18 // 3)
```

8. Which of the following operators is used for identity comparison in Python?

- a) == b) = c) is d) in

9. What will be the output of the following Python code?

```
my_dict = {"name": "John", "age": 25, "city": "Mumbai"}
print(my_dict.get("salary", "Not Found"))
```

- a) None b) 0 c) Not Found d) Error

10. What possible output is expected from the following code?

```
import random
L = [5, 15, 25, 35]
lower = random.randint(1, 1)
upper = random.randint(2, 3)
for k in range(lower, upper + 1):
    print(L[k], end="@")
```

- a) 15@25@ b) 5@15@ c) 15@25@35@ d) 25@35@

11. What will be the output of the following Python code?

```
i = 10
print(i, end="--")
def change():
    global i
    i = i + 5
    print(i, end="###")
change()
print(i)
```

- a) 10--15###15 b) 10--10###15 c) 10--15###10 d) 15--15###15

12. What is the output of the given Python code?

```
text = 'Hello-World-Python'
print(text.split("-"))
```

- a) ['Hello', 'World', 'Python'] b) ['Hello-World-Python']
c) ['Hello', 'World-Python'] d) Error

13. Which of the following will correctly import only the sqrt and ceil functions from the math module?

- a) import math b) from math import *
c) from math import sqrt, ceil d) import sqrt, ceil from math

14. **Assertion (A):** The expression (1, 2, 3).append(4) will generate an error.

Reason (R): Tuples are immutable data types in Python.

Choose the correct option:

- a) Both A and R are True and R is the correct explanation for A
b) Both A and R are True but R is not the correct explanation for A

c) A is True but R is False

d) A is False but R is True

15. Which of the following is used to take input from the user in Python?

a) print()

b) input()

c) output()

d) scan()

16. State whether the following statement is **True** or **False**:

The len() function in Python returns the number of characters in a string including spaces.

17. What will be the output of the following Python code?

```
a = 10
```

```
b = 20
```

```
print(a > b or b > a)
```

a) True

b) False

c) Error

d) No output

SECTION B (5 × 2 = 10 Marks)

18. A. Explain the difference between **break** and **continue** statements in Python with a suitable example.

OR

B. Explain the difference between **explicit** and **implicit** type conversion in Python with a suitable example.

19. The code below is intended to remove the first character and the last character of a string and return the resulting string. Rewrite it after correcting all errors. Underline the corrections made.

```
def modify_string(str)
    if len(str) < 2:
        return str
    new_str = str[0:-1]
    return new_str
result = modify_string("Python")
Print("Modified string: " result)
```

20. A. (Using Python built-in methods/functions only):

- I. Write a statement to convert the string "hello world" to upper case.
- II. Write a statement to find the index of the first occurrence of the substring "is" in a string named sentence.

OR

B. Predict the output of the following Python code:

```
text = "Learn Python programming with fun"
print(text.partition("with"))
print(text.count("\n"))
```

21. A. Write a function remove_item() in Python that accepts a list L and a number n. If the number exists in the list, remove its first occurrence. If it does not exist, print "Element not found".

OR

B. Write a Python function add_student() that accepts a dictionary students, a roll_no, and a name. The function should add the roll number and name to the dictionary. If the roll number already exists, print "Student already exists" instead of updating it.

22. Differentiate between:

- I. Local variable and Global variable
- II. list() and tuple() functions

SECTION C (2 × 3 = 6 Marks)

23. Write a Python function that displays the number of times the word "**the**" appears in a text file named "story.txt". Assume each word is separated by spaces.

24. A. Predict the output of the following Python code:

```
s1 = "CODE-123"
s2 = ""
i = 0
while i < len(s1):
    if s1[i] >= '0' and s1[i] <= '9':
        num = int(s1[i])
        num = num + 1
        s2 = s2 + str(num)
    elif s1[i] >= 'A' and s1[i] <= 'Z':
        s2 = s2 + s1[i+1] if i+1 < len(s1) else s1[i]
    else:
        s2 = s2 + '*'
    i += 1
print(s2)
```

OR

B. Predict the output of the following Python code:

```
cities = ["Mumbai", "Delhi", "Bangalore", "Chennai", "Kolkata"]
result = []
for city in cities:
    if city[0] in 'ABCDE':
        result.append(city.upper())
print(result)
```

SECTION D (3 × 4 = 12 Marks)

25. Write a Python program to perform the following operations on a list of numbers:

- **I.** Create a list of 10 numbers (you may hardcode or take input).
- **II.** Find and print the **maximum**, **minimum**, and **sum** of the numbers in the list.
- **III.** Count and print the **frequency** of a specific number (ask the user which number to search for) using linear search.

Example:

If the list is [10, 20, 30, 20, 40, 20] and the user searches for 20, the output should be Frequency of 20 is 3.

26. A. Write a Python program to:

- Create a dictionary with names of 5 employees as keys and their salaries as values.
- Display all employees whose salary is greater than 50000.
- Find the total salary of all employees.

OR

B. Write a Python program to:

- Create a list of 10 numbers entered by the user.
- Find and display the maximum, minimum, and sum of all numbers in the list.
- Also display the numbers in sorted order.

27. Predict the output of the following Python code. Write the output for each print statement:

Code 1

```
L1 = [1, 2, 3, 4, 5]
L1.append([6, 7])
print(len(L1))
```

Code 2

```
tup = (10, 20, 30, 40, 50)
print(tup[1:4])
```

Code 3

```
d = {'a': 1, 'b': 2, 'c': 3}
d['d'] = 4
print(d.keys())
```

Code 4

```
for i in range(2, 10, 2):
    if i == 6:
        continue
    print(i, end=" ")
```

SECTION E (1 × 5 = 5 Marks)

28. A. Write a Python program using a user-defined function that accepts a list of numbers as an argument and returns the sum of all even numbers in the list. Also write the main program that:

1. Takes 5 numbers from the user and stores them in a list.
2. Calls the function with this list.
3. Prints the sum of even numbers.

Example:

If the list is [3, 6, 7, 8, 11], the output should be Sum of even numbers: 14 (because 6 + 8 = 14).

OR

B. Write a Python program that:

1. Takes a string input from the user.
2. Counts and displays the number of vowels (a, e, i, o, u - both uppercase and lowercase) in the string.
3. Also displays the number of consonants in the string.

Example:

Input: "Hello"

Output: Vowels: 2, Consonants: 3